

```
/* =====  
/* ARRAYS IN JAVASCRIPT  
/* =====
```

/* JavaScript Array is a data structure that allows you to store and organize multiple values within a

/* It can hold various data types, including numbers, strings, objects, and even other arrays.
/* Arrays in JavaScript are zero-indexed i.e. the first element is accessed with an index 0, the second with index 1, and so on.

```
/* =====  
/* Creating Arrays:  
/* =====
```

```
//? Using Array constructor  
// let fruits = new Array('apple', 'orange', 'banana')
```

```
//? Using array literal  
// let fruits = ["apple", "orange", "banana"];  
// console.log(fruits);
```

```
/* =====  
/* Accessing Elements:  
/* =====
```

```
//? Accessing Elements: Array elements are accessed using zero-based indices.  
// let fruits = ["apple", "orange", "banana"];  
// console.log(fruits[2]);  
// console.log(fruits[3]);
```

```
/* =====  
/* Modifying Elements:  
/* =====
```

//? Modifying Elements: You can modify array elements by assigning new values to specific indices

```
// let fruits = ["apple", "orange", "banana"];  
// fruits[2] = "mango";  
// console.log(fruits);
```

```
/* =====  
/* Array Traversal:  
/* =====
```

```
// let fruits = ["apple", "orange", "mango", "grapes", "banana"];
```

```
//? 1: for of loop  
/* for...of Loop: The for...of loop is used to iterate over the values of arrays, strings
```

```
// for (let item of fruits) {  
//   console.log(item);  
// }
```

```
// for (let item = 0; item < fruits.length; item++) {  
//   console.log(fruits[item]);  
// }
```

```
//? 2: for in loop  
/* for...in Loop: The for...in loop is used to iterate over the indices of an array, string.
```

```
// for (let item in fruits) {  
//   console.log(item);  
// }
```

// Template string

```
// let firstName = "jatin";  
// let LastName = "Bhosale";
```

```
// console.log(firstName + LastName);  
// console.log(`${firstName} ${LastName}`);
```

```
// let age = 20;  
// console.log(`I am ${age} years old`);
```

//? 3: for each method

```
// fruits.forEach((curElem, index) => {  
//   console.log(`${curElem} ${index}`);  
// });
```

//? 4.map function

```
// const map = fruits.map((curElem, index) => {  
//   return `${curElem} ${index}`;  
// });  
// console.log(map);
```